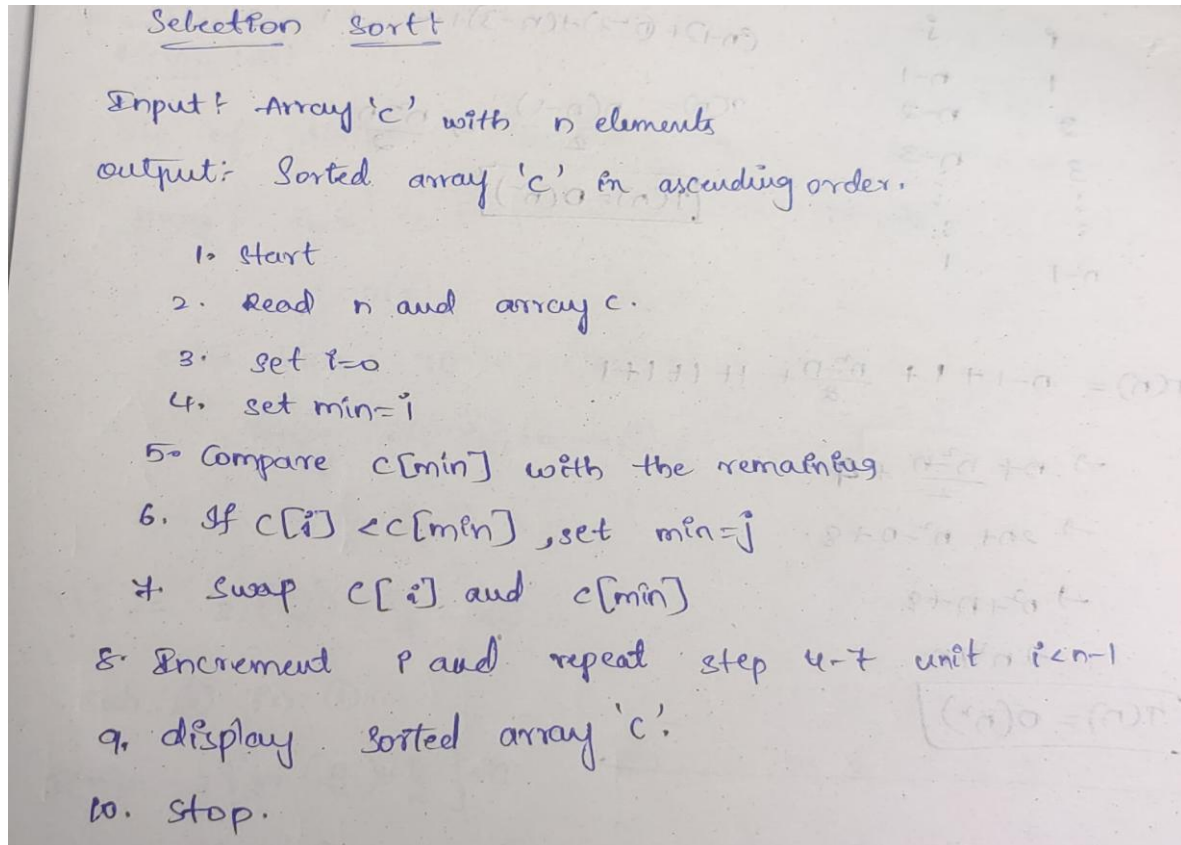


Aim: To Implementation and time analysis of selection sort algorithm.

Algorithm for Selection sort:



Program of Selection sort:

```
#include <iostream>
using namespace std;

class SelectionSortApp
{
public:
    int arr[100], n;

    void arrayInput()
    {
        cout << "Enter number of elements: ";
        cin >> n;
```

```
    cout << "Enter the elements: ";
    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }
}

void printArray()
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}

// Selection Sort Function
void sort()
{
    for (int i = 0; i < n - 1; i++)
    {
        int minIndex = i;

        for (int j = i + 1; j < n; j++)
        {
            if (arr[j] < arr[minIndex])
            {
                minIndex = j;
            }
        }
        if (minIndex != i)
        {
            int temp = arr[i];
            arr[i] = arr[minIndex];
            arr[minIndex] = temp;
        }
    }
}
};
```

```
int main()
{
    SelectionSortApp ss;
    ss.arrayInput();
    cout << "\nArray before sorting: ";
    ss.printArray();
    ss.sort();
    cout << "\nArray after Selection Sort: ";
    ss.printArray();

    return 0;
}
```

Output:

Output

```
Enter number of elements: 5
Enter the elements: 2
1
8
4
3

Array before sorting: 2 1 8 4 3

Array after Selection Sort: 1 2 3 4 8

=== Code Execution Successful ===
```

Time complexity:

Time Complexity:-

```

for ( ) {          → n
    minIndex = i;   → 1
    for ( ) {       → n
        if ( ) {     → 1
            // " " → 1
        }
        if ( ) {
            // " " → 1
        }
    }
}

```

~~$T(n) = n + 1 + n + 1 + 1$~~
 $T(n) = (n+1)(n+1+1)$
 $= (n+1)(n+2)$
 $= n^2 + 2n + n + 2$
 $T(n) = n^2$
 $O(n^2)$